CHAPTER 1: REVIEW AND ADVANCED NETWORK FUNDAMENTALS

Types of Networks

1. Personal Area Network (PAN)

The smallest and most basic network type. It is designed for a single person, typically within a range of **10 meters**.

- **Purpose:** Syncing personal devices like smartphones, tablets, and laptops.
- **Common Tech:** Bluetooth, ZigBee, and USB.
- **Example:** Connecting your wireless earbuds to your phone or a smartwatch to a tablet.

2. Local Area Network (LAN)

A LAN connects devices within a limited area such as a single home, office, or school building. It is characterized by high data transfer speeds and low latency.

- **Purpose:** Sharing resources like printers, files, and internet access among a group of users.
- **Common Tech:** Ethernet (IEEE 802.3) and Wi-Fi (IEEE 802.11).
- **Example:** A home Wi-Fi network or a computer lab in a school.

3. Campus Area Network (CAN)

A CAN is an interconnection of multiple LANs within a specific geographical area, such as a university campus or a large medical complex.

- **Purpose:** Linking various departments (e.g., the library to the registrar's office) while remaining under the control of a single entity.
- **Example:** A university network where the Engineering building and the Arts building are on different LANs but connected via a campus-wide backbone.

4. Metropolitan Area Network (MAN)

A MAN spans an entire city or a large campus. It is larger than a LAN but smaller than a WAN, typically covering **5 to 50 kilometers**.

- **Purpose:** Connecting multiple LANs across a city to provide high-speed connectivity to local government offices or residential areas.
- **Common Tech:** Fiber-optic cables and high-capacity wireless links.
- **Example:** A city-wide cable TV network or a municipal "Smart City" network.

5. Wide Area Network (WAN)

A WAN covers a massive geographical area, such as a country, a continent, or the entire globe.

- **Purpose:** Connecting distant LANs and MANs so that a user in one location can communicate with a user thousands of miles away.

- **Example:** The **Internet** is the largest and most well-known public WAN. Private WANs are used by multinational corporations to connect global branch offices.

Specialized Network Types

Beyond geographical scale, some networks are defined by their specific function:

- **Storage Area Network (SAN):** A specialized, high-speed network that provides block-level network access to storage. It connects servers to data storage devices (like tape libraries or disk arrays).
- **Virtual Private Network (VPN):** Not a physical network, but a secure "tunnel" created over a public network (like the Internet) to allow remote users to access a private network securely.
- **Wireless Local Area Network (WLAN):** A LAN that uses wireless communication (Wi-Fi) instead of cables to link devices.

Data Representation

Data is a piece of information that can be analog or digital. It can be in the form of a number, character, text, audio, or video. It can be created, deleted, stored, or transferred. It has two (2) forms:

- **Analog data** refers to continuous information.
    - Example: human voice
- **Digital data** takes on discrete values.
    - These bits can be a file, information, or instruction.
    - These consist of ones and zeros (1's and 0's).

Signal Forms

- Periodic signal — It completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods. The completion of one (1) full pattern is called a cycle.
- Nonperiodic signal — It changes without exhibiting a pattern or cycle that repeats over time. Both analog and digital signals can be periodic or nonperiodic.

Parts of Signals

1. Peak

- **Definition:** The highest point of a wave above its average (rest) position.
- **Meaning:** Represents the maximum positive value of the signal.
- **Example:** In a sound wave, the peak is where air molecules are most compressed.

2. Wavelength (λ)

- **Definition:** The distance between two identical points in consecutive cycles of a wave (e.g., peak-to-peak or trough-to-trough).
- **Meaning:** It tells how "long" one cycle of the wave is in space.
- **Relation:** $\lambda = \frac{\text{Wave speed}}{\text{Frequency}}$
- **Example:** In light waves, red light has a longer wavelength than blue light.

3. Amplitude

- **Definition:** The maximum displacement of the wave from its average (rest) position.
- **Meaning:** Represents the strength or intensity of the signal — higher amplitude means stronger signal.
- **Example:** In audio, higher amplitude means louder sound.

4. Trough

- **Definition:** The lowest point of a wave below its average (rest) position.
- **Meaning:** Represents the maximum negative value of the signal.
- **Example:** In a water wave, the trough is the deepest part between two crests.

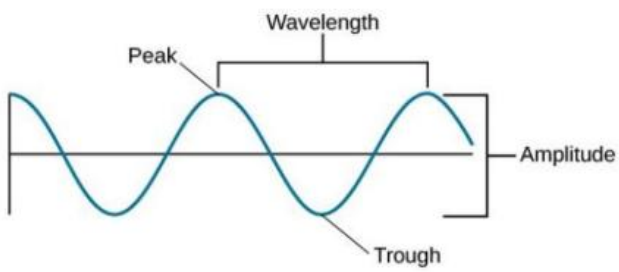


Figure 1. Parts of a Signal
Source: Fundamentals of Data Communication Networks, 2017

Channel Impairments

When a signal travels from the **transmitter** to the **receiver**, it's never "perfect." The journey affects its **strength**, **quality**, and **shape**. These changes are called **impairments**, and they happen in three main forms:

1. Attenuation – The **weakening** of a signal's strength as it travels. If the signal gets too weak, it becomes **indistinguishable from noise**.

2. Noise – Any **unwanted signal** that mixes with the desired signal. Reduces clarity and may cause errors in digital communication.

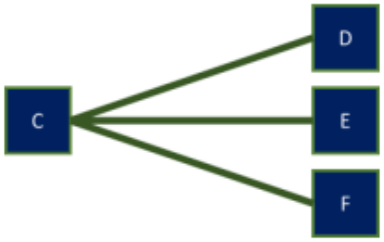
3. Distortion – The **change in shape or form** of the signal as it travels. Different frequency components of a signal travel at different speeds through the medium.

Network Topologies

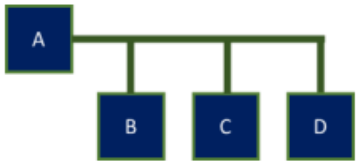
These are the physical or logical layouts of devices that belong in a network.

- **Point-to-Point (P2P) topology** — It is a link that permanently connects two (2) nodes or network devices.

- **Point-to-Multipoint topology** — One (1) node is connected to multiple nodes, each in a P2P manner.

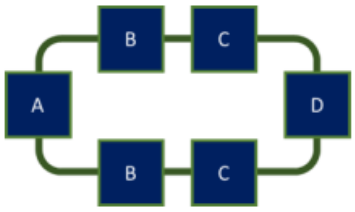


- **Multidrop topology** — All nodes are interconnected by a single link with one (1) node that is the master node and the other nodes are secondary or slave nodes.

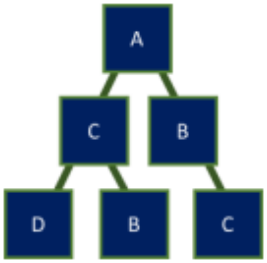


- **Bus topology** — It is like the multidrop topology with the exception that there is no master—slave relationship; all nodes are peers. The line terminator is used to prevent a signal that comes to the end of a transmission line from bouncing back and corrupting other signals on the line.

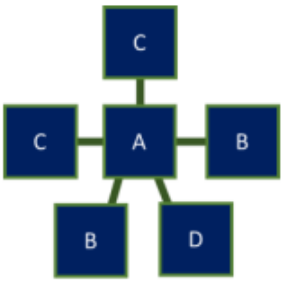
- **Ring topology** — The nodes are connected serially in a P2P manner withs the last node connected to the first node to form a loop.



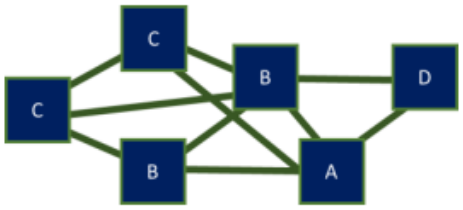
- **Tree topology** — It is formed by connecting multiple buses to form a system of branching links with no closed loop.



- **Star topology** — It is a topology in which each node is connected in a P2P manner to a central node called a hub.



- **Mesh topology** — The network nodes are interconnected arbitrarily. Generally, users are connected to only a subset of the nodes, and another set of internal nodes provides a switching facility that moves data from one node to another until it reaches its destination.



Network Design Principles

When moving into advanced networking, we follow the Cisco Hierarchical Design Model. Instead of a "flat" network where everyone is plugged into one giant switch, we break it into three layers to improve management and speed:

- **Access Layer:** This is the "front door." It's where end-user devices (PCs, printers, IP phones) connect to the network.
- **Distribution Layer:** This acts as the "bridge." It routes data between different local networks (VLANs), implements security policies, and aggregates the connections from the access switches.
- **Core Layer:** The "high-speed backbone." Its only job is to switch traffic as fast as humanly possible between distribution layer devices. It shouldn't be bogged down by security rules or user traffic.

While the **Hierarchical Model** (Access, Distribution, Core) focuses on how traffic flows, the **Enterprise Network Model** (specifically the Cisco Enterprise Architecture) describes the functional "blocks" of a large organization.

Think of it as the difference between the **skeleton** (hierarchy) and the **rooms in a building** (enterprise modules). Each "room" or module has a specific job and can be scaled or secured without breaking the others.

Enterprise Network Model

The Enterprise Model breaks a large network into four main areas:

- **Enterprise Campus:**
  - This is where the users live. It includes the Access, Distribution, and Core layers we discussed. It also houses the **Internal Server Farm** (DNS, email, file servers) and network management tools.
  - This is the "internal" world where the company's employees and data reside. It usually follows the 3-layer hierarchical structure (Access, Distribution, Core).
- **Enterprise Edge:**
  - This is the "security perimeter." It acts as the gateway to the outside world. This is the most critical part of the architecture. It is the interface between the internal campus and the untrusted outside world. It includes:
    - **E-commerce module:** For customer-facing web servers.
    - **Internet module:** Providing general web access for employees.
    - **VPN/Remote Access:** For workers logging in from home.
    - **WAN module:** To connect the main office to other branches.
- **Service Provider (SP) Edge**
  - This represents the external world that you don't control, such as your Internet Service Provider (ISP) or Public Switched Telephone Network (PSTN).

- This layer isn't owned by the company; it represents the external networks that allow you to reach other locations.

- **Remote Modules**
  - These are the "outposts" of the network, like **Enterprise Branches** (satellite offices) or **Teleworkers** (individual remote employees).
    - **A. Enterprise Branch (The Satellite Office) –** This is a permanent remote location, like a local bank branch, a retail store, or a regional sales office.
      - **What's inside:** A small-scale version of the campus (Access and Distribution layers), local IP phones, and a **Branch Router**.
      - **Connectivity:** Usually connects via a dedicated WAN link (like MPLS) or a Site-to-Site VPN.
    - **B. Enterprise Teleworker (The Home Office) –** This represents individual employees working from home or a coffee shop.
      - **What's inside:** A laptop, a small SOHO (Small Office/Home Office) router, and a software VPN client.
      - **Connectivity:** Uses the public Internet to create a secure "tunnel" to the Enterprise Edge.

References:

- Forouzan, B. A. (2017). *Data Communications and Networking* (5th ed.). McGraw-Hill Education.
- Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach* (8th ed.). Pearson Education.